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Gatsby Computational Neuroscience Unit, UCL
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WORK EXPERIENCE & EDUCATION

06|2026 Postdoctoral Research Fellow

Gatsby Computational Neuroscience Unit, Sainsbury Wellcome Center,
University College London, United Kingdom.
Primary collaborators: Dr. Agostina Palmigiano.

05|2026 Postdoctoral Research Fellow

Computational and Biological Learning Lab, Department of Engineering,
University of Cambridge, United Kingdom.
Primary collaborators: Prof. Máté Lengyel, Dr. Yashar Ahmadian.
from 03|2023 Royal Society Newton International Fellow

05|2022 Fast track Ph.D. in Computational Neuroscience

“Plasticity of Inhibition in Recurrent Circuits.” Goethe University, Frankfurt am Main, Germany.
from 09|2019 Max Planck Institute for Brain Research, Advisor: Prof. Julijana Gjorgjieva.
from 09|2015 Frankfurt Institute for Advanced Studies, Advisor: Prof. Jochen Triesch.

04|2015 Bachelor of Science, Physics

“Cubic Learning Rules for Unsupervised Self-Limiting Hebbian Learning in Artificial Neural Networks.”
Goethe University, Frankfurt am Main, Germany.

09|2014 Term Abroad at the University of Birmingham

Courses in Psychology and Computer Science. University of Birmingham, Birmingham, United Kingdom.

ONGOING PROJECTS

2026 “Plasticity and function of hippocampal E-I memory networks,”
S. Eckmann, Y. Ahmadian, M. Lengyel. – [Conference abstract] [Conference abstract] [COSYNE talk]

2026 “Structured E-I networks for movement generation”
S. Eckmann, L. Xi-Zi, G. Hennequin, Y. Ahmadian, M. Lengyel. – [Conference abstract]

2026 “Homeostatic regulation of neural activity affords efficient sensory coding”
E. J. Young, Y. Ahmadian, S. Eckmann. – Senior author [Thesis Chapter 4]

2026 “Multi-subtype inhibitory networks from synapse-type-specific competitive learning”
R. I. Ianov, Y. Ahmadian, M. Lengyel, S. Eckmann. – Senior author [Conference abstract]

JOURNAL PAPERS

2024 “Synapse-type-specific competitive Hebbian learning forms functional recurrent networks,”
S. Eckmann, E. J. Young, J. Gjorgjieva, **PNAS**. – [Paper]

2020 “Active Efficient Coding Explains the Development of Binocular Vision and its Failure in Amblyopia,”
S. Eckmann, L. Klimmasch, B. E. Shi, J. Triesch, **PNAS**. – [Paper]

2015 “The Fisher Information as a Neural Guiding Principle for Independent Component Analysis,”
R. Echeveste, S. Eckmann, C. Gros, **Entropy**. – [Paper]

GRANTS & AWARDS

03|2026 COSYNE Presenter Travel Award

Granted to attend the COSYNE conference 2026 in Lisbon, Portugal.

03|2023 Royal Society Newton International Fellowship

Personal grant by The Royal Society to study “Inhibition stabilized hippocampal circuits.”
To be held for three years at the University of Cambridge.

- 03|2022 NAISYS Travel Award**
Granted in support of attending the NAISYS conference 2022 in Cold Spring Harbour, NY, USA.
- 06|2019 C3N Summer School**
Cellular, Computational and Cognitive Neuroscience, Princeton, New Jersey, USA.
- 02|2019 COSYNE New Attendee Travel Award**
Granted to attend the COSYNE conference 2019 in Lisbon, Portugal.
- 08|2018 Logistics of Neuronal Function Summer School**
Giersch International Symposium & Summer School, Frankfurt am Main, Germany.
- 09|2016 Visual Neuroscience Summer School**
From Spikes to Awareness, Schloss Rauischholzhausen, Germany.
- 09|2014 Goethe University Strategic Partnership Program**
Scholarship granted by the German Academic Exchange Service (DAAD) to study one term in the UK.

INVITED & CONTRIBUTED TALKS

- 04|2026 Gatsby Computational Neuroscience Unit, University College London, hosted by Agostina Palmigiano**
“Anti-Hebbian plasticity protects against memory interference,” London, UK.
- 03|2026 COSYNE – Computational and Systems Neuroscience Conference**
“Anti-Hebbian plasticity protects against memory interference,” Lisbon, Portugal.
- 03|2026 BonnBrain - Young Investigator Symposium**
“Anti-Hebbian plasticity protects against memory interference,” Bonn, Germany.
- 07|2025 Gordon Research Conference on Inhibition in the Central Nervous System**
“Normalization of hippocampal activity affords theta-separated memory encoding and recall,” ME, USA.
- 07|2025 Center for Theoretical Neuroscience, Columbia University, hosted by Ken Miller**
“Theta-modulated memory encoding and retrieval in recurrent hippocampal circuits,” New York, USA.
- 09|2024 Bernstein Computational Neuroscience Conference**
“Theta-modulated memory encoding and retrieval in recurrent hippocampal circuits,” Frankfurt, Germany.
- 07|2024 UKNC – UK Neural Computation Conference**
“A neural circuit model for inhibition-stabilized memory encoding and retrieval,” Sheffield, UK.
- 06|2024 Junior Scientists Workshop on Recent Advances in Theoretical Neuroscience**
“Structured inhibition-stabilized supralinear networks,” Trieste, Italy.
- 05|2024 Cambridge Memory Meeting**
“A neural circuit model for theta-modulated memory encoding and retrieval,” Cambridge, UK
- 03|2024 Champalimaud Centre for the Unknown, Petreanu Lab, Group Meeting**
“Top-down modulated surround suppression,” Lisbon, Portugal.
- 10|2021 Search Symposium, Cognitive Science Department, Osnabrück, Germany**
“Computation and learning in biological neural networks,” Osnabrück, Germany.
- 05|2021 Bernstein SmartSteps Seminar Series**
“A theory for Hebbian Learning in recurrent E-I networks,” online.
- 12|2020 Computational and Biological Learning Lab, Group Meeting, Department of Engineering**
“A theory for Hebbian Learning in recurrent E-I networks,” Cambridge, UK.
- 11|2020 Research Institute for Signals, Systems and Computational Intelligence – sinc(i)**
“A theory for Hebbian Learning in recurrent E-I networks,” Santa Fe, Argentina.
- 09|2020 Max Planck Institute for Brain Research, Institute Seminar**
“A theory for Hebbian Learning in recurrent E-I networks,” Frankfurt am Main, Germany.
- 08|2018 Computational and Mathematical Models in Vision**
“An active efficient coding model of the development of amblyopia,” St. Pete Beach, Florida, USA.

CONFERENCE POSTERS

- 03|2026 BonnBrain**
“Anti-Hebbian plasticity protects against memory interference,” S. Eckmann*, M. Cooper*, M. Moniri, M. Abuelem, D. M. Bannerman, E. O. Mann, Y. Achmadian, M. Lengyel. Bonn, Germany.
- 07|2025 UKNC – UK Neural Computation Conference**
“Hebbian construction of inhibition-stabilized networks with multiple inhibitory neuron subtypes supports contextual modulation,” R. I. Vitanov, S. Eckmann. London, UK.
- 07|2025 GRC & S – Gordon Research Conference & Seminar on Inhibition in the Central Nervous System**
“Normalization of hippocampal activity affords theta-separated memory encoding and recall,” S. Eckmann, Y. Ahmadian, M. Lengyel. Newry, Maine, USA.
- 02|2024 COSYNE – Computational and Systems Neuroscience Conference**
“Inhibition-stabilized supralinear memory ensembles,” S. Eckmann, Y. Ahmadian, M. Lengyel. Lisbon, Portugal.
- 02|2023 COSYNE – Computational and Systems Neuroscience Conference**
“Input-dominated Hebbian learning enables image-computable E-I networks,” S. Eckmann, Y. Ahmadian, M. Lengyel. Montreal, Canada.
- 08|2022 EVCM – European Visual Cortex Meeting**
“Synapse-type-specific competitive Hebbian learning forms functional recurrent networks,” S. Eckmann, J. Gjorgjieva. Seeon, Germany.
- 04|2022 NAISYS – From Neuroscience to Artificially Intelligent Systems Conference**
“Unsupervised competitive Hebbian learning explains the emergence of functional recurrent E-I networks,” S. Eckmann, J. Gjorgjieva. Cold Spring Harbor, New York, USA.
- 02|2021 COSYNE – Computational and Systems Neuroscience Conference**
“A theory for Hebbian plasticity in recurrent E-I networks,” S. Eckmann, J. Gjorgjieva. Online.
- 09|2020 Bernstein Computational Neuroscience Conference**
“Hebbian learning of stable receptive fields in recurrent E-I networks,” S. Eckmann, J. Gjorgjieva. Online.
- 02|2019 COSYNE – Computational and Systems Neuroscience Conference**
“Stable memories despite large spontaneous synaptic fluctuations,” S. Eckmann, S. S. Jhutti, J. Triesch. Lisbon, Portugal.
- 08|2018 ECVF – European Conference on Visual Perception**
“A computational model of the development and treatment of anisometropic amblyopia,” S. Eckmann, L. Klimmasch, B. E. Shi, J. Triesch. Trieste, Italy.
- 05|2018 VSS – Vision Science Society Annual Conference**
“A model of the development of anisometropic amblyopia through recruitment of interocular suppression,” S. Eckmann, L. Klimmasch, B. E. Shi, J. Triesch. St. Pete Beach, Florida, USA.
- 05|2017 VSS – Vision Science Society Annual Conference**
“A computational model for the joint development of accommodation and vergence control,” J. Triesch, S. Eckmann, B. E. Shi. St. Pete Beach, Florida, USA.
- 07|2015 CNS – Annual Computational Neuroscience Meeting**
“Should Hebbian learning be selective for negative excess kurtosis?” C. Gros, S. Eckmann, R. Echeveste. Prague, Czech Republic.
- 06|2015 EITN – European Institute for Theoretical Neuroscience Workshop on Learning and Plasticity**
“An Objective Function for Hebbian self-stabilizing Plasticity Rules,” R. Echeveste, S. Eckmann, C. Gros. Paris, France.
- 05|2015 OCCAM – Osnabrück Computational Cognition Alliance Meeting**
“From Stationarity to ICA: an Objective Function for Hebbian self-stabilizing Plasticity Rules,” R. Echeveste, S. Eckmann, C. Gros. Osnabrück, Germany.

TEACHING EXPERIENCE

- 2023-26 Teaching Assistant in Computational Neuroscience** – *Undergraduate & graduate level*
Grading and correction of course assignments.
- 2017 Teaching Assistant in Theoretical Neuroscience** – *Undergraduate & graduate level*
Design, correction, and presentation of exercises. Grading of final exams.
- 2013 Teaching Assistant in Theoretical Physics** – *Undergraduate level*
Conducting tutorials in theoretical electrodynamics. Presentation of exercises. Grading of final exams.

SERVICE & LEADERSHIP

- 2026** Bernstein workshop organizer: “Plasticity and function of hippocampal memory circuits.”
- 2026** Reviewer for Nature Communications & Science Advances
- 2025** Discussion leader – Gordon Research Seminar on Inhibition in the Central Nervous System.
- 2024** Reviewer for the Wellcome Early-Career Award.
- 2024** Participant in In2ScienceUK outreach event at The Royal Society.
- 2023/24** Reviewer for COSYNE – Computational and Systems Neuroscience Conference.
- 2023** COSYNE workshop organiser: “Shaping circuit functions via plastic and diverse inhibition.”
- 2022** Participant in the “Ideation Workshop on Neuromorphic Computing“ by the Volkswagen Foundation.
- 2021** Co-initiator of the cross-disciplinary initiative “Learning in Spiking Neural Networks” at Goethe University.
- 2018** Initiator and organiser of the “Comp. Neuro. Journal Club” at the Frankfurt Institute for Advanced Studies.

MENTORING

- 2026** **Rebeca Ianov Vitanov**: “Functional models of cortical circuits.” – *PhD*
- 2025** **Edward James Young**: “Homeostatic scaling in recurrent neural networks.” – *PhD*
- 2023** **Mete Hergul**: “Normalization in recurrent networks.” – *Thesis project, afterwards hedge fund analyst*
- 2022** **Abraham Alsawaf**: “Homeostatic scaling in recurrent E-I networks.” – *Thesis project, afterwards MD student*
- 2020** **Ashish Rao Mangalore**: “Noise quenching in plastic Stabilized Supralinear Networks” – *Internship*
- 2019** **Melanie Tinz**: “Stochastic trafficking of synaptic resources affect spine lifetimes” – *Internship*
- 2019** **Suneet Singh Jhutti**: “Neuronal balance through homeostasis.” – *Thesis project, afterwards PhD student*
- 2019** **Nils Möbus**: “An introduction to Principal Component Analysis.” – *Thesis project*